

Van Tien Pham

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Education

PhD in Computer Science

11/2022 – 11/2025

Université de Toulon, France

Thesis: Deep Neural Network Compression using Pruning and Low-rank Approximations.

Supervisors:

Thanh Phuong Nguyen, Full Professor	Université Côte d’Azur	Supervisor
Yassine Zniyed, Associate Professor	Université de Toulon	Co-supervisor

Jury:

Nicolas Gillis, Full Professor	Université de Mons	Reviewer
Jenny Benois-Pineau, Full Professor	Université de Bordeaux	Reviewer
Vicente Zarzoso, Full Professor	Université Côte d’Azur	Examiner
Mohammed Nabil El Korso, Full Professor	Université Paris-Saclay	Examiner
Alice Caplier, Full Professor	Université Grenoble Alpes	Examiner
Thanh Phuong Nguyen, Full Professor	Université Côte d’Azur	Supervisor
Yassine Zniyed, Associate Professor	Université de Toulon	Co-supervisor

Master’s Degree in Computer Science

10/2019 – 12/2020

Hanoi University of Science and Technology (HUST), Vietnam

Thesis: Object detection, segmentation, and tracking from egocentric vision.

GPA: 4.0/4.0. Classification: Very good.

Engineering Degree in Computer Science

09/2014 – 06/2019

Hanoi University of Science and Technology (HUST), Vietnam

Programme de Formation d’Ingénieurs d’Excellence au Vietnam (PFIEV).

GPA: 3.2/4.0. Classification: Very good.

Professional Experience

Postdoctoral Fellow

12/2025 – Present

Laboratoire d’Informatique et des Systèmes (LIS), UMR 7020 CNRS, France

Research on efficient multimodal foundation models for anomaly detection.

Research Engineer

07/2019 – 10/2022

Viettel High Technologies Industry Corporation, Viettel Group, Vietnam

Development of a driving simulator using 3D simulation engines for training applications, integrating computer vision, virtual reality, and facial recognition modules to enhance user interaction and realism.

Intern in Computer Vision

08/2018 – 02/2021

MICA International Research Institute, HUST – CNRS/UMI-2954 – Grenoble INP, Vietnam

Research on image classification, object detection, segmentation, and tracking using neural networks.

Publications

Journals

- [J5] [Van Tien Pham](#), Yassine Zniyed, Thanh Phuong Nguyen, **Coupled tensor decomposition for compact network representation**, IEEE Transactions on Neural Networks and Learning Systems, vol. 37, no. 2, pp. 617-631, February 2026.
- [J4] [Van Tien Pham](#), Yassine Zniyed, Thanh Phuong Nguyen, **Singular values-driven automated filter pruning**, Neural Networks, vol. 192, p. 107857, December 2025.

- [J3] Neriman Tokcan*, Shakir Showkat Sofi*, Van Tien Pham*, Clémence Prévost*, Sofiane Kharbech*, Baptiste Magnier, Thanh Phuong Nguyen, Yassine Zniyed, Lieven De Lathauwer, **Tensor decompositions for signal processing: theory, advances, and applications**, Signal Processing, vol. 238, p. 110191, January 2026. *These authors contributed equally to this work as co-first authors.
- [J2] Van Tien Pham, Yassine Zniyed, Thanh Phuong Nguyen, **Enhanced network compression through tensor decompositions and pruning**, IEEE Transactions on Neural Networks and Learning Systems, vol. 36, no. 3, pp. 4358–4370, March 2025.
- [J1] Van Tien Pham, Yassine Zniyed, Thanh Phuong Nguyen, **Efficient tensor decomposition-based filter pruning**, Neural Networks, vol. 178, p. 106393, October 2024.

Preprints

- [P1] Van Tien Pham, Yassine Zniyed, Thanh Phuong Nguyen, **Decoupling matrix-valued functions using zeroth- and first-order information and its application in neural network compression**, 2025.

Conferences

- [C6] Van Tien Pham, Yassine Zniyed, Thanh Phuong Nguyen, **Hybrid network compression through tensor decompositions and pruning**, 32nd European Signal Processing Conference (EU-SIPCO), Lyon, France, pp. 1052–1056, August 2024.
- [C5] Van Tien Pham, Yassine Zniyed, Thanh Phuong Nguyen, **Élagage efficace des filtres basé sur les décompositions tensorielles**, XXIXème Colloque Francophone de Traitement du Signal et des Images (GRETSI), Grenoble, France, pp. 937-940, August 2023.
- [C4] Van Tien Pham, Thanh Phuong Nguyen, **Identification and localization of COVID-19 abnormalities on chest radiographs**, The International Conference on Artificial Intelligence and Computer Vision, pp. 251-261, 2023.
- [C3] Van Tien Pham, Cong Minh Tran, Stanley Zheng, Tri Minh Vu, Shantanu Nath, **Chest X-ray abnormalities localization via ensemble of CNNs**, International Conference on Advanced Technologies for Communications, 2021.
- [C2] Van Tien Pham, T.H. Tran, H. Vu, **Detection and tracking hand from FPV: benchmarks and challenges on rehabilitation exercises dataset**, IEEE International Conference on Computing and Communication Technologies (RIVF), 2021.
- [C1] Van Tien Pham, Thi Lan Le, Thanh Hai Tran, Thanh Phuong Nguyen, **Hand detection and segmentation using multimodal information from Kinect**, IEEE International Conference on Multimedia Analysis and Pattern Recognition (MAPR), 2020.

Reviewer

Journals: IEEE Transactions on Pattern Analysis and Machine Intelligence, Expert Systems with Applications, Neural Networks, Neurocomputing, Engineering Applications of Artificial Intelligence, Quantum Machine Intelligence, Mathematics, Electronics, Applied Sciences, The Journal of Supercomputing, Scientific Reports, Signal, Image and Video Processing, njp Heritage Science, Journal of King Saud University Computer and Information Sciences.

Conferences: ICLR, AISTATS, IJCNN, ICMI, GRETSI.

Skills

Programming languages: Python, C, R, Pascal, L^AT_EX, HTML

Frameworks and tools: PyTorch, Keras, Slurm (Jean Zay HPC), Matlab, Qt, Unreal Engine, OpenCV, Git, GitHub, MySQL, PostgreSQL

Spoken languages: English (Professional proficiency), French (Intermediate), Vietnamese (Native)

Honors

2021: Best Paper Runner-Up Award, IEEE International Conference on Computing and Communication Technologies (RIVF)

2020: First-Class Honours, Master's Degree, Hanoi University of Science and Technology